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Seasonal differences in the spatial patterns of wildfire drivers and susceptibility in the southwest mountains of China

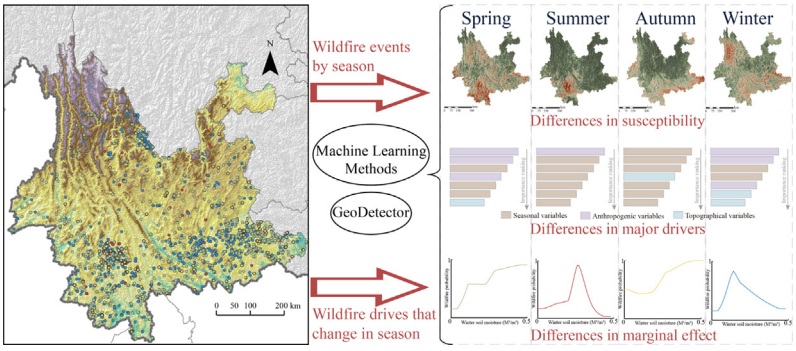
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HIGHLIGHTS

- Integrate machine learning and statistical models to analyze wildfire susceptibility
- The uniqueness of seasonal wildfire patterns facilitates differential strategies.
- Different drivers dominate wildfire occurrence in fire and nonfire seasons.
- The wildfires' response to drivers across seasons improves the understanding of lagged effects.

GRAPHICAL ABSTRACT



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ABSTRACT

Wildfires directly affect global ecosystem stability and severely threaten human life. The mountainous areas of Southwest China experience frequent wildfires. Mapping the susceptibility patterns and analyzing the drivers of wildfires are crucial for effective wildfire management, especially considering that the inclusion of seasonal dimensions will produce more dynamic results. Using Yunnan Province of China as a case study area, a method was attempted to distinguish dependable wildfires by season, while possible wildfire drivers were obtained and refined within seasons. The patterns of wildfire susceptibility in different seasons were modeled based on the Maxent and random forest models. Then, the spatial relationships between wildfire and potential drivers were analyzed integrating with GeoDetector to evaluate the variable importance of drivers and the marginal effect of drivers. The results showed that the two models effectively depicted each season's wildfire susceptibility. The susceptible wildfire areas in spring and winter are located throughout Yunnan Province, with anthropogenic factors being the most significant drivers. During the summer and autumn, wildfire risk areas are relatively concentrated, showing a trend of dominant drought-driven and humid conditions. The differences in wildfire drivers across seasons reflect the lagged effect of climate factors on wildfires, leading to significant discrepancies in the marginal effects of how seasonal drivers affect wildfires. The findings improve our understanding of the effects of the interseasonal variability of environmental variables on wildfires and promote the development of specific seasonal wildfire management strategies.

1. Introduction

Despite wildfires are a natural phenomenon of ecosystems (McWethy et al., 2019), uncontrolled wildfires have become a major environmental

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problem at both global and local scales. They have destroyed vegetation ecosystems (Vukomanovic and Steelman, 2019), posing a continuing threat to soil composition (Dove et al., 2020; Jimenez-Morillo et al., 2020), water balance (Venkatesh et al., 2020), air quality (Sokolik et al., 2019), life and biodiversity (Haifeng et al., 2021; Jaafari et al., 2017). Wildfires frequently occur in China, with an average of 12,683 wildfires/year and an annual affected area of 6748 km^2, which is greater than the world average (Xie et al., 2022). Notably, the mountainous region in Southwest China around the Indo-China Peninsula is considered a global wildfire hotspot and ranks as a high-risk area for wildfires (Bowman et al., 2009). The heterogeneity of climate, vegetation, and topography associated with mountainous terrain creates many challenges for local wildfire management (Cui et al., 2022; Shangqi et al., 2021) and fire suppression personnel (Hayes, 2020; Xiong et al., 2020). In this context, a better understanding of the unique local wildfire susceptibility patterns and drivers becomes critical to wildfire management. Furthermore, proper wildfire management is important for protecting forest ecosystems and maintaining biodiversity in mountainous areas (Bassett et al., 2017).

Wildfire occurrence is related to a combination of factors, such as ignition sources and the conditions that promote the spread of fire in specific temporal and spatial patterns (Jaafari et al., 2017; Oliveira et al., 2012). These factors, which include meteorological conditions, vegetation, topography, and anthropogenic variables, can vary between seasons (e.g., seasonal changes in temperature and precipitation) and largely influence the occurrence and development of wildfires (Matin et al., 2017; Zeng et al., 2022). Thus, wildfire susceptibility patterns show obvious interseasonal variability (Littell et al., 2009; Sturtevant et al., 2009). In the mountains of southwestern China, wildfires are generally regulated by monsoons from the western Pacific and Indian Oceans (Ying et al., 2022). The period when the monsoon recedes and is replaced by dry southwestern air currents mostly coincides with the wildfire season (i.e., from December to the following May). During this period, wildfire management was focused on establishing strict fire prevention measures (e.g., blocking forest areas). However, due to the limited number of local forestry personnel and the financial limitations that impact wildfire management, it is difficult to achieve continuous and uniform management across the landscape (Xiong et al., 2020). Assessing interseasonal wildfire susceptibility can promote dynamic adaptation among preventive measures and landscape management and land use planning practices (Trucchia et al., 2022).

On a seasonal scale, climate change usually affects wildfires by changing the moisture content of fuels through temperature variability, precipitation, and evapotranspiration or by affecting vegetation, thus altering the composition of fuel loads (Badia, 2002; Roads et al., 2005). However, such effects may have a lagged response, i.e., wildfire occurrence in a particular season is often influenced by the cumulative effect of the environmental factors from the previous season (Keeley and Syphard, 2016). For example, the effect of rainfall on wildfires during a season may not be immediately apparent due to the interception and storage of moisture by the forest canopy. This cumulative effect may lead to the same causative factor influencing wildfire occurrence to different extents across seasons. Such an example is shown in northwestern Italy, where the same wildfire driver (the maximum temperature of the warmest month) strongly impacts summer wildfires while not affecting winter wildfires (Vacchiano et al., 2018). Another case is depicted in California, where the quantified importance of wildfire drivers varies obviously across seasons (Qiu et al., 2022). Those interseasonal drivers of wildfire in the mountains of southwestern China have not yet been fully explored. Previous research mainly assessed annual wildfire susceptibility without considering the temporal characteristics of drivers over the year and identified the discrepancies of critical drivers in this area (Lan et al., 2021; Ye et al., 2017; Zhang et al., 2019; Zhu et al., 2022), which may be attributed to wildfires responding to distinct drivers in different temporal dimension. When focusing on the dynamics of drivers, the contribution to wildfire distribution may vary over time (Rodrigues et al., 2016; Sun et al., 2021). As mentioned above, there are lagging and cumulative effects of wildfire drivers. Therefore, considering the seasonal variety of wildfire drivers can improve our understanding of the impact of interseasonal variation on wildfire occurrence. This is crucial to explain wildfire-driving mechanisms and predict future wildfire occurrence. Furthermore, the framework fully accounting for seasonal changes in drivers to assess wildfire susceptibility across time needs to be described explicitly at a global scale.

In this study, we model wildfire susceptibility over four seasons in Yunnan Province, a typical mountainous region in Southwest China integrating machine learning and statistical approaches. Various wildfire drivers are considered including static and dynamic (climate, vegetation and their seasonal differences) drivers. The objectives of this work were to assess wildfire susceptibility in each season and explore the importance of wildfire drivers at a seasonal scale by distinguishing the seasonal variations in wildfire and drivers. In addition, we determined whether same drivers similarly influence wildfire occurrence across seasons. In the process, we intentionally compared the random forest with the Maxent in modeling wildfire susceptibility. Ultimately, we aim to better understand the mechanisms that drive short-term wildfires and provide a novel perspective for dynamic wildfire susceptibility studies. Locally, we hope to provide decision-makers with appropriate guidance and early warning to improve wildfire management strategies in different seasons for mountainous southwest China with unique climatic and landscape heterogeneity.

2. Materials and methods

2.1. Study area

The study area was Yunnan Province, a region with typical mountainous topography on the southwest border of China, located on the Yunnan-Guizhou Plateau between 21°09′–29°15′N and 97°30′–106°12′E (Fig. 1). The terrain presents a step-like distribution that is high in the northwest and low in the southeast. The plateau and mountains occupy >90 % of the region, and the scattered basins comprise <10 % of the land. The highest elevation in the province is 6740 m, and the lowest is only 76.4 m. Yunnan has a highland tropical monsoon climate, controlled by warm and humid tropical monsoon air masses in summer and dry and hot continental air in winter, with simultaneous rain and heat. The significant altitude differences and the complex topographic landscape result in the uneven distribution of the local climates. These different microclimates make Yunnan Province an extremely biologically rich region, ranking among the 25 global biodiversity hotspots (Yang et al., 2004). The biological diversity, unique climatic conditions and local slash and burn agricultural practices make Yunnan Province one of the most severe wildfire regions (Ying et al., 2021); thus, this location is a typical wildfire study area from the mountainous region of Southwest China. In addition, the occurrence of wildfires has a strong seasonal pattern. Fires are mainly concentrated in winter and spring (from December to May of the following year), while the other two seasons have low occurrence but continue to threaten the local area (Fig. 1a).

2.2. Wildfire historical dataset

Although the wildfire occurrence data collected by government departments have more explicit temporal and spatial information, such data are usually not easily accessible, and the data integrity is limited. In contrast, remote sensing data obtained by the Moderate Resolution Imaging Spectrometer (MODIS) radiometer Terra and Aqua satellites, which offer comprehensive spatiotemporal coverage and freely available data, have been used as a suitable and reliable source for monitoring wildfires (Hantson et al., 2013). The data used to represent historical fire points are the MODIS Active Fire Products dataset from the Fire Information for Resource Management System (FIRMS). As verified by Giglio et al. (2016) and Hantson et al. (2013), this dataset typically has a false alarm rate of <10 % and can better identify wildfires occurring in small areas compared to other wildfire products. In general, the dataset can detect flames and smoky fires over an area of up to 1000 m^2. Under favorable observation conditions, fires of one-tenth of this size can be detected (Zhuang et al.,

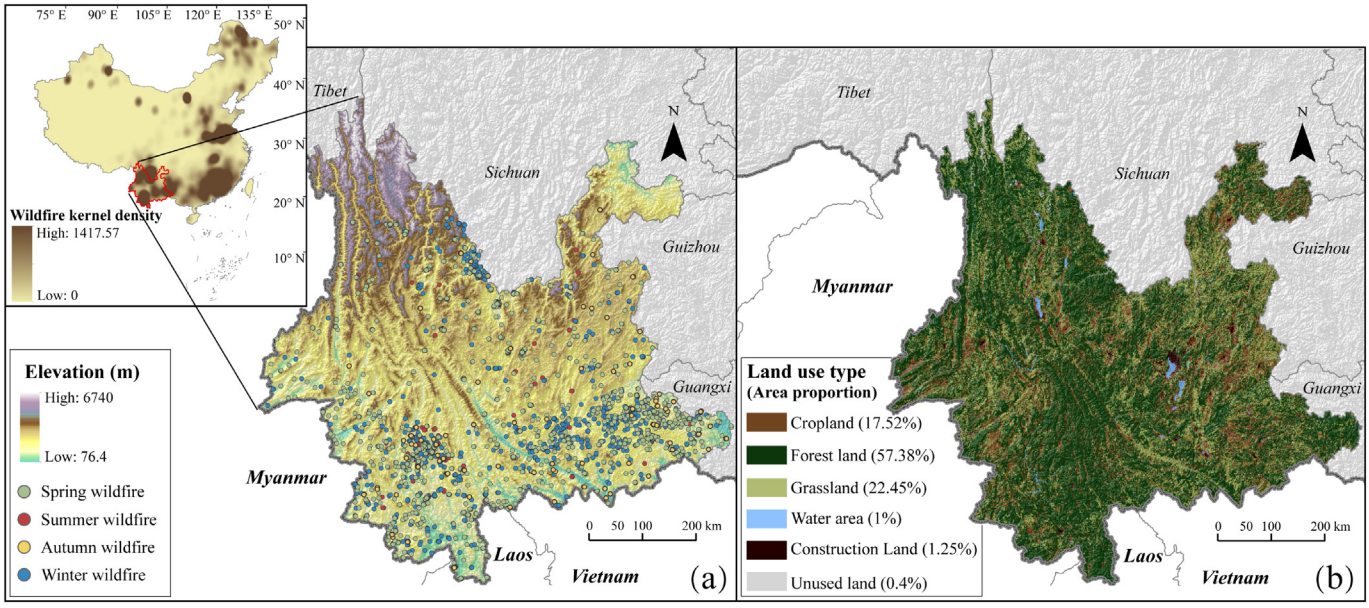


Fig. 1. (a) Location of the study area and distribution of MODIS active fire events in 2020 (wildfire events come from the FIRMS dataset with >80 % confidence). National scale maps show the local wildfire kernel density for the corresponding period. (b) The land use types and area proportion in 2020.

2018). Each point in the data represents one or more fires that may exist within a 500 m radius of that point. Among these points, we considered only the points of presumed vegetation fire (type field is 0). In addition, the data record information such as quality assurance metrics for each scanned pixel and fire radiation power (FRP), which indicates the intensity of the fire. Fire points with a confidence interval of <80 % were removed to obtain highly dependable fire points. Furthermore, to avoid the incorrect inclusion of active fire sources, those agricultural straw burning and fires that occurred within agricultural and developed land (based on raster-categorized land use types) were excluded (Fig. 1b). By processing the data in the above way, false alarms for nonfire points were avoided as much as possible. Temporal patterns of wildfire occurrence and average intensity (reflected by FRP) were counted for different seasons during the study period (2006–2020).

2.3. Wildfire drivers

2.3.1. Seasonal wildfire drivers

Climate and vegetation factors were considered seasonal wildfire drivers in this study due to their significant differences across seasons (Bajocco et al., 2017; Littell et al., 2009; Qiu et al., 2022). The meteorological data for this study were primarily obtained from the National Earth

Table 1
List of wildfire drivers used in this study (each seasonal variable contains four subvariables for different seasons).

Category	Type	Indicator (abbreviation)/(unit)	Resolution	Time period	Data source
Seasonal wildfire drivers	Climate	Temperature (TEM_SPR, TEM_SUM, TEM_AUT, TEM_WIN)/(0.1 °C)	1 km	2006–2020	National Earth System Science Data Center (http://www.geodata.cn)
		Accumulated precipitation (PRE_SPR, PRE_SUM, PRE_AUT, PRE_WIN)/(mm)	1 km	2006–2020	
		Wind speed (WND_SPR, WND_SUM, WND_AUT, WND_WIN)/(m/s)	1 km	2006–2020	
		Potential evapotranspiration (PET_SPR, PET_SUM, PET_AUT, PET_WIN)/(0.1 mm)	1 km	2006–2020	
		Relative humidity (RH_SPR, RH_SUM, RH_AUT, RH_WIN)/(%)	1 km	2006–2020	
		Vapor pressure difference (VPD_SPR. VPD_SUM, VPD_AUT, VPD_WIN)/(kPa)	1 km	2006–2020	
		Soil moisture (SM_SPR, SM_SUM, SM_AUT, SM_WIN)/(M^3/m^3)	1 km	2006–2019	National Tibetan Plateau Data Center (http://data.tpdc.ac.cn)
	Vegetation	Normalized difference vegetation index (NDVI_SPR, NDVI_SUM, NDVI_AUT, NDVI_WIN)	1 km	2006–2020	National Earth System Science Data Center (http://www.geodata.cn)
Nonseasonal wildfire drivers	Topography	Elevation (ELE)/(m)	90 m		Digital Elevation Model data from Resource and Environment Science and Data Center (http://www.resdc.cn)
		Aspect index (ASPECT)	90 m		
		Slope (SLOPE)/(°)	90 m		
		Topographic wetness index (TWI)	90 m		
	Anthropogenic factors	Proximity to road (Prox to Road)/(m)	N/A	2020	OpenStreetMap (https://www.openstreetmap.org)
		Proximity to farmland (Prox to Farmland)/(m)	1 km	2020	Resource and Environment Science and Data Center (http://www.resdc.cn)
		Gross domestic product (GDP)/(million yuan/km^2)	1 km	2005, 2010, 2015, 2019	
		Population density (POP)/(people/km^2)	1 km	2006–2020	WorldPop (https://hub.worldpop.org)

Notes: According to Ma et al. (2022), the aspect is cosine transformed to obtain a linear index (aspect index) that better distinguishes the degree of exposure. In addition, the TWI is calculated with reference to Raduła et al. (2018). In this study, spring (referred to as SPR) includes March–April–May, summer (SUM) includes June–July–August, autumn (AUT) includes September–October–November, and winter (WIN) includes December–January–February.

System Science Data Sharing Infrastructure. This dataset included monthly observations of mean temperature, cumulative precipitation, relative humidity, potential evapotranspiration, and mean wind speed. As another driver, vapor pressure deficit (VPD) alters vegetation water stress, so we estimated the VPD based on Gharun et al. (2015). In addition, we obtained monthly data on soil moisture from the National Tibetan Plateau Data Center, which potentially affects vegetation moisture (Rigden et al., 2020). To explore the effect that the intrayear variability of the seasonal drivers has on wildfires across seasons, we subdivided each driver by season and extracted the values for each season separately (Table 1).

Similarly, vegetation factors that differed across seasons were considered for the normalized difference vegetation index (NDVI), which is the most commonly used indicator of vegetation cover and growth status, to represent the combustible load (Huang et al., 2019). We obtained the maximum NDVI values from the raster in each season to construct the four subvariables. The NDVI data were downloaded from the National Earth System Science Data Center, which is monthly data synthesized from the MODIS MOD13A2 product. All these variables were averaged separately for the quarterly values from 2006 to 2020 to portray long-term quarterly differences in the seasonal wildfire drivers.

2.3.2. Nonseasonal wildfire drivers

Although topographic factors do not vary seasonally, the important role of topographic factors in influencing the occurrence and spread of seasonal wildfires still cannot be ignored (Bjånes et al., 2021), especially in mountainous terrain. Altitude, slope, and aspect can affect the amount and structure of vegetation combustibles, as well as the possibility of anthropogenic fire sources, which are closely related to the occurrence of wildfires (He et al., 2022). The topographic wetness index (TWI) reflects the ability of the landscape to accumulate water and moisture (Raduła et al., 2018). The elevation, slope, aspect index and TWI within each grid in the study were derived from the digital elevation model (DEM) dataset and used directly in the modeling process.

With the increasing impact of human activities on wildfires, it has become increasingly necessary for anthropogenic factors to be considered as wildfire drivers (Ye et al., 2017). The anthropogenic factors in this study were primarily socioeconomic related and specifically included the gross domestic product (GDP), population density, proximity to farmland, and proximity to roads. The gridded GDP data were obtained from the Resource and Environmental Science and Data Center, and annual values were averaged for 2005, 2010, 2015, and 2019 based on the chronological range of the study. The population density was averaged for 2006–2020 using the World Population website. The distance of each grid from the road is the Euclidean distance calculated from the actual roads acquired in the Open Street Map (OSM). As a substantial open access global spatial database, OSM has higher positioning accuracy than other publicly available datasets (Kolanek et al., 2021). In addition, due to the consideration that wildfires are frequently caused by agricultural-related human activities, the farmland raster was extracted from the land use type data in 2020, and then the Euclidean distance of each grid from the farmland was calculated. A total of 40 wildfire drivers were included in the study after unification into a 1-km resolution considering the assessment of climate, vegetation, topography, anthropogenic variables and those seasonalities. The data sources and interpretation of each wildfire driver are provided in Table 1.

2.4. Initial exploration of the drivers

2.4.1. GeoDetector identification

One of the focuses of this study was to explore the relationship between wildfire occurrence and drivers across seasons. Therefore, we first resorted to the GeoDetector method to determine the relative contribution of wildfire drivers. GeoDetector is a statistical method proposed by Wang et al. (2010) that uses four detectors (factor, interaction, risk, and ecological) to detect spatial differentiation and reveal the driving forces of that differentiation. Unlike traditional regression analysis, GeoDetector has no linear assumptions and therefore does not need to consider multicollinearity among drivers (Wang and Xu, 2017). The general assumption of applying GeoDetector in wildfire research can be expressed as follows: if a driver contributes to the occurrence of wildfire, the spatial distribution of wildfire should be similar to the spatial distribution of that driver. This study used the factor detection function of GeoDetector to explain the relative importance of wildfire drivers, and the degree of explanation was measured by the q-value, calculated as:

$$q = 1 - \frac{\sum_{h=1}^{L} N_h \sigma_h^2}{N\sigma^2} \tag{1}$$

where $h = 1, \ldots, L$ is the strata of variable Y or factor X; N_h and N are the strata h and the number of units in the whole area; σ_h^2 and σ^2 are the variances of the Y value of strata h and the whole area, respectively. The value range of q is [0.1], and a larger value of q indicates a stronger determination ability or relative importance of the independent factor X on the attribute Y. Interaction detector was also used in the study to check whether the combination of the two variables enhances or weakens their effect on wildfire. As well as risk detector was used to examine the magnitude of differences in the mean attribute values of the drivers differs across subzones (Venkatesh et al., 2022). We used the "GD" package in the R programming environment to implement the GeoDetector function (Song et al., 2020).

2.4.2. Premodeling driver screening

It is difficult to avoid the problem of multicollinearity because the same class of drivers in different seasons may be highly correlated or non-independent. Although GeoDetector does not have to consider this problem, this can introduce instability when using subsequent machine learning modeling (Dormann et al., 2013). We therefore calculated the Pearson and Spearman correlation coefficients between each driver and eliminated one from each pair based on the any of the correlation coefficients ($|r| > 0.75$ or $|rho| > 0.75$) (Zhang et al., 2018). The specific criterion was that, based on the q-value (derived from the GeoDetector) of each driver, when the correlation coefficient exceeded the threshold, the driver with a lower q-value between the two was eliminated. Thereafter, the datasets may still have a missing correlation structure following the rejection of these drivers (Bektas et al., 2022). To address this, we performed an additional variance inflation factor (VIF) test after correlation assessment to ensure that the modeled drivers did not have potential multicollinearity problems (VIF < 10) (Dormann et al., 2013; Hong et al., 2018).

2.5. Machine learning modeling

2.5.1. Maxent model

The Maxent model is a machine learning technique for density estimation and event distribution prediction based on maximum entropy theory, which is constrained to the environmental variables at each event point (i.e., only the points that exist) and estimates the probability distribution of the target by iteratively searching for the probability distribution with maximum entropy (Phillips et al., 2006). This method was initially applied with modeling species distributions from multidimensional environmental inputs and has been extended to the study of spatial patterns of wildfire susceptibility in recent years with desirable results (Carlson et al., 2021; Parisien et al., 2016; Yang et al., 2021). The Maxent model uses a pure presence framework, which is essential compared to the presence-absence framework, because the locations of truly nonexistent wildfires may be unknown. To improve the confidence of sample points that indicate the "presence" of wildfires, we adopted a new fire point sample treatment strategy for the fire season (spring and winter), i.e., only rasters with more than two wildfires over 15 years within a 1 km range were used as wildfire sample points. However, this strategy was not applied to samples from the nonfire season (summer and autumn), and only multiple wildfire points within a raster were removed to balance the total number of samples in different seasons. We repeatedly screened these wildfire sites to obtain modeled wildfire samples for four seasons: spring (868), summer (310), autumn (150), and winter (380).

The accuracy of Maxent models cannot be accurately estimated, so appropriate optimization measures are required to limit model complexity and overfitting (Phillips et al., 2006). The best performing model is usually determined by controlling two parameters that affect the model performance: 1) the regularization multiplier (RM) and 2) the feature combination (FC). In this study, we used the "Kuenm" R package (Cobos et al., 2019) to generate all possible 29 feature class combinations and 10 regularization multipliers (from 0.5 to 5 in increments of 0.5) for each of the four seasons. A total of 290 models were generated per season, and the performance of the candidate models was evaluated by statistical significance (via partial receiver operating characteristic curves, or ROC), predictive ability (omission rate), and model complexity (using corrected Akaike information criterion, or AICc, values). The final models selected were statistically significant, possessed low omission rates (e.g., <5 % when possible) and had the smallest delta AICc value (Cobos et al., 2019).

2.5.2. Random forest model

Random forest (RF) is an ensemble method based on classification or regression trees (Leo, 2001). The model works by creating multiple decision trees with randomly selected bootstrap samples to obtain a prediction for each tree. Each tree's split is determined using a random subset of drivers for each node. The final result is a combination (majority or average) of the outcomes of all trees (Valavi et al., 2021). This algorithm overcomes the instability problem of using a single classification tree, resulting in higher prediction accuracy and offering superior results in wildfire susceptibility modeling (Gholamnia et al., 2020; Iban and Sekertekin, 2022). Unlike Maxent's pure presence framework, RF modeling requires presence and absence samples, which are not available. This problem was solved by creating a pseudoabsence sample. We created pseudoabsence points with the same number of fires present at random in a region 10 km away from all fire points (Barbet-Massin et al., 2012).

To obtain the best output, we tuned the ntree (number of trees in forest) and mtry (number of variables randomly sampled at each split) parameters in the RF model (Pahlavan-Rad and Akbarimoghaddam, 2018). By iterating the mtry value from 1 to 20 and setting five ntree values of 200, 400, 600, 800, and 1000, we found the optimal combination with the highest accuracy. The modeling process we repeated the 10-fold cross-validation 10 times, and the dataset was randomly divided into new folds before each repetition to improve the robustness of the models. These processes were implemented with the "randomForest" R package (Liaw and Wiener, 2002).

2.6. Susceptibility mapping and performance evaluation

We randomly divided wildfire sample points (the RF model also includes pseudoabsence points) into a training set containing 75 % of all points and a test set that contained the remaining 25 % (Zhao et al., 2022). Model tuning and construction were obtained from the training set, and the test set was used to verify the performance of both models in each season.

Model construction was based on raster maps of all variables to generate wildfire susceptibility maps. These susceptibility maps depicted the suitability of a given pixel for ignition, relative to all other pixels (Bar Massada et al., 2013). Pixel values ranged from 0 to 1. In addition, we superimposed the raster with pixel values >0.5 in each season to derive a high-risk season number map of Yunnan Province. We evaluated the accuracy and fit of the models using the area under the curve (AUC) value of the ROC curve, which is a valid threshold independent metric and can be considered as the probability that a randomly present sample is correctly predicted (Merow et al., 2013). An AUC value of 0.5 indicates that the model does not perform better than random, while a value closer to 1.0 indicates better model performance (Merow et al., 2013). Models with AUC values higher than 0.75 are usually considered valid (Phillips and Dudík, 2008).

2.7. Driver importance analysis

In addition, we assessed the importance of drivers that control the spatial pattern of wildfire probabilities across seasons using two approaches. First, we used a combination of each model's native metrics (mean decrease accuracy, or MDA, for RF and permutation importance for Maxent) and the q-values of GeoDetector. This was done by calculating the average importance rank of each driver across models and creating a composite ranking of the importance (Syphard and Franklin, 2009). In addition, we calculated the AUC values of the model using a single variable in turn, which helped understand the variable importance of the drivers.

Eventually, we further analyzed the partial dependency plots (i.e., marginal effects) of a single variable by predicting each response using one variable at a time when other variables were held constant (i.e., mean value) to investigate the direction of influence of each variable. Similarly, we determined the optimal extent of the partial variable impact on wildfire occurrence through risk detection.

3. Results

3.1. Ignition/wildfire driver selection

A total of 16,178 wildfire sites meeting the confidence level were identified over 15 years, most of which were concentrated in the spring (10,131) and winter (5574) seasons, while the number of summer (322) and autumn (151) seasons was quite low (Fig. 2). Spring wildfires were the dominant wildfire composition in all years except for 2009, 2011, 2017 and 2018, when winter wildfires accounted for the largest number. These four years outputted the least/or lesser total number of wildfires. Despite slight variations, the spring wildfires provided the greatest FRP each year by examining the intensity. The difference was that wildfires with higher FRP values occurred in the winter of 2006 and 2013 as well as during the autumn of 2018 (which was the highest FRP that year).

The correlation test of the variables revealed that seasonal drivers in some different seasons had strong correlations, while correlations among

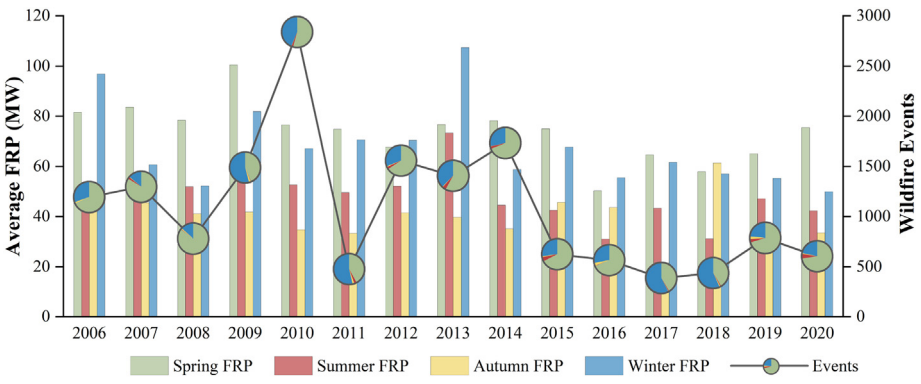


Fig. 2. Temporal distribution of average intensity and number of events for MODIS active fire. The pie chart shows the ratio of wildfire events by season.

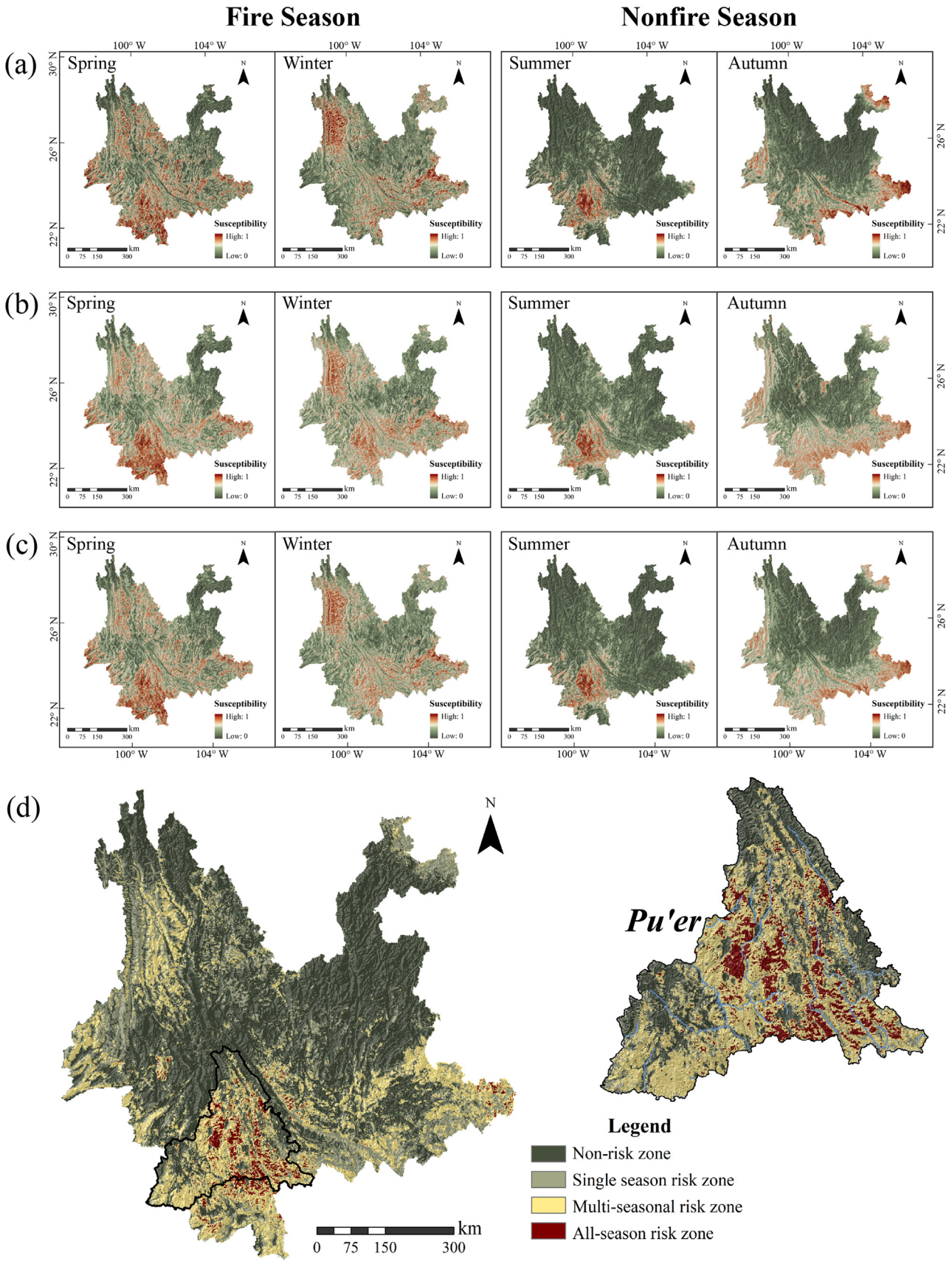


Fig. 3. Wildfire susceptibility map: (a) Maxent modeling results, (b) RF modeling results, (c) weighted average results, (d) high-risk season number map (showing which areas are threatened by a sustained high probability of wildfires over multiple seasons).

variables other than climatic factors were generally weak (Fig. S1). Although there were classes of drivers with strong correlations across seasons, they had different data distribution characteristics (Fig. S2). We drew on the q-value output in the GeoDetector as a basis for determining the drivers to be excluded in the final modeling. We found in the factor detection results that the dry season (winter and spring) climate had a more prominent effect on each season, which showed the highest or high q-value for wildfires in different seasons. In contrast, summer climate factors rarely had the highest q-values across seasons. The exception is that summer precipitation is significantly more critical for wildfires in all seasons (Table S1). Interaction detection showed the combined effect of the two drivers on wildfire occurrence, with each set of driver interactions having a stronger effect on wildfire than a single one, and this enhancement was more of a nonlinear-enhance (Fig. S3). The drivers with lower q-values of the pair of drivers whose correlation exceeded the threshold were deleted, and the final number of factors retained for each season is detailed in Table S2. Subsequent multicollinearity tests ensured that the VIF values of the factors were below 10 in the wildfire models for each season (Table S2).

3.2. Model results and performance evaluation

Regarding the optimization of the Maxent models, the 290 candidate Maxent models generated in each season were statistically significant. Among these models, at least one satisfying the omission rate criterion (<5 %) was present during spring, summer and winter. From these, we selected models for each season that met the omission rate while having the lowest delta AICc value, namely, spring (RM = 2, FC = PTH), summer (RM = 2.5, FC = QT), and winter (RM = 1.5, FC = LQP) (Fig. S4). Among the candidates for autumn, none of the models met the omission rate criterion, and we chose the combination of the lowest omission rate and delta AICc value (RM = 1.5, FC = LQ). Among the candidate RF models, we selected the parameter combinations with the highest accuracy to construct the models, which were spring (mtry = 2, ntree = 1000), summer (mtry = 5, ntree = 600), autumn (mtry = 5, ntree = 400) and winter (mtry = 2, ntree = 600) (Fig. S5).

After tuning, both the Maxent models and the RF models achieved good prediction results. The AUC values of all models were >0.8, except for the autumn model constructed with Maxent, which had an AUC value slightly lower than 0.8 (Fig. S6). The performance of the two methods was almost equal, but the RF model always had a higher AUC value and better prediction accuracy than the Maxent model.

3.3. Mapping of wildfire susceptibility patterns by season

The wildfire susceptibility obtained by the two methods was almost identical in terms of the spatial pattern (Fig. 3a, b). The magnitude of the AUC values was used as a weight to average the two methods, and final susceptibility maps were obtained (Fig. 3c). For the fire season, the areas with wildfire probabilities higher than 0.5 in spring and winter accounted for 30.09 % and 25.74 % of the areas under the jurisdiction of Yunnan Province, respectively (Fig. 3). During the fire season, a considerable part of Yunnan Province is threatened by a high probability of wildfires. In terms of spatial distribution patterns, there were significant geographic differences in wildfire risk during the fire season. Areas with high wildfire probability in spring were scattered throughout the province, especially in the south, west, and northwest regions (Fig. 3c). The common feature of these areas is the distribution of large areas of forest that maintain a relatively intact natural environment. In particular, the near-tropical areas located in the south had a significantly concentrated and extreme probability of spring wildfires. This region has continually preserved the highest forest cover in the province and is also the region that has been subject to the highest number of wildfires. In contrast, the overall wildfire probability was low in the northeast. This area is known as a dry and hot valley region where climatic conditions are favorable for wildfire formation but also where the distribution of combustible vegetation is sparse. Therefore, this region is not at a high risk of spring wildfires. Winter wildfire susceptibility patterns were slightly more concentrated in the northwest, southeast, and east regions than in spring (Fig. 3c). Unlike the high wildfire probability in spring in the southern region, the winter wildfire probability was significantly lower in the southern region, indicating seasonal

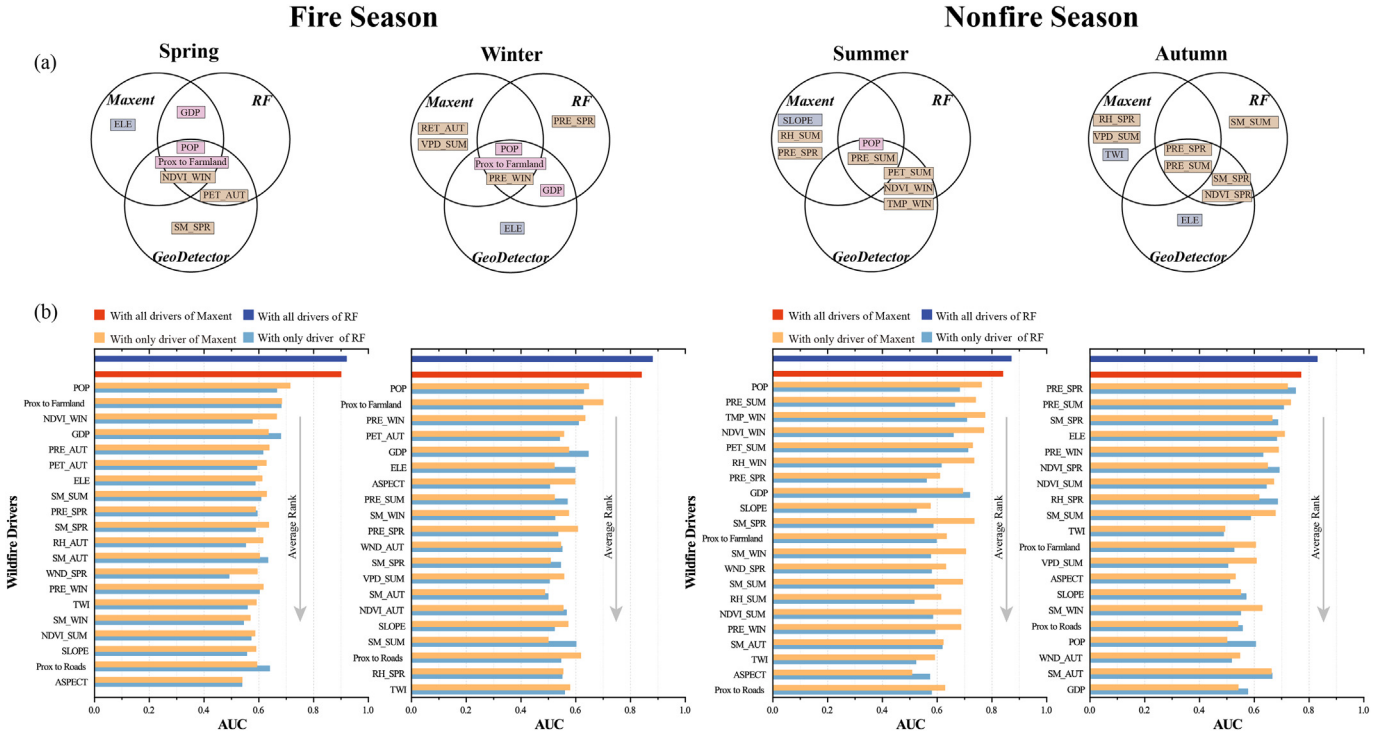


Fig. 4. In the fire seasons and nonfire seasons, (a) the top five important drivers obtained by the three methods (the colors distinguishing the driver are seasonal driver, anthropogenic factor, or topographic factor) and (b) AUC values obtained in turn using single variable modeling. Abbreviations same as in Table 1.

differences in wildfire occurrence in this region. In addition, the winter wildfire risk in the northwestern region was significantly higher than that in spring. This region is the highest elevation area in Yunnan Province, with treacherous mountains and canyons. The probability of these combustible materials burning is higher in winter when the temperature is low, but the local vegetation cover is dense.

For the nonfire seasons, there were fewer areas with wildfire probabilities above 0.5, accounting for 10.74 % (summer) and 22.61 % (autumn) of the total area of Yunnan Province. Although wildfires less threatened most areas in both seasons, these areas exhibited significant seasonal differences in spatial patterns (Fig. 3). Areas of high wildfire risk in summer were concentrated in the southern part of Yunnan, which was also an area of very high wildfire risk in spring. The wildfire risk in summer was not as high as that in spring in the southernmost rainforest regions. This may be due to the high summer humidity in the region, which is not conducive to wildfire formation. In autumn, the high wildfire probability area was distributed around the southeastern provincial border, which is dominated by low elevation hilly terrain and is another area in Yunnan with high dense forest areas. This area is the closest part of Yunnan Province to the ocean, and the adjacent coastal province of Guangxi has suffered from a high incidence of wildfires in the autumn for a long time.

Fig. 3d shows that almost all of Yunnan Province has high wildfire risk in at least one season to a greater or lesser extent. The city of Pu'er in the south faces a high risk of wildfire threats throughout the year. In contrast, a small area in the northeast is relatively safe, with a consistently low wildfire risk across all seasons.

3.4. Variable importance of drivers by season

For spring wildfires, all three native indicators suggested that population (POP) and proximity (Prox) to farmland had high relative importance (Fig. 4a). Similarly, for summer wildfires, the comparatively vital drivers were POP and summer precipitation (PRE_SUM) (Fig. 4a). PRE_SUM and spring precipitation (PRE_SPR) were essential in autumn, while POP, Prox to farmland, and winter precipitation (PRE_WIN) were important in winter (Fig. 4a). After combining the importance rankings of the three primary variables, the variable importance of the drivers that influence wildfires in different seasons was revealed (Table 2). Based on the average ranking, anthropogenic factors (POP and Prox to farmland) reflected the highest importance in both fire seasons. No other anthropogenic factors proved significant for the nonfire seasons except for a strong association between wildfires and POP in summer. During these two seasons, wildfires are unfamiliar, while the key factors triggering wildfires depend on climatic factors, especially summer precipitation (PRE_SUM) (Table 3). Unexpectedly, only one climatic variable (autumn precipitation, PET_AUT) was among the most important drivers affecting spring wildfires, and the others were all anthropogenic and vegetation related. The situation was similar in winter, but winter precipitation (PRE_WIN) also played a vital role in winter wildfire occurrence. In summer and autumn, anthropogenic factors have become almost irrelevant to the occurrence of wildfires because of the unfavorable burning environment. Therefore, the climate conditions impacting the burning environment (including PRE_SUM, TMP_WIN, PET_SUM, PRE_SPR, PRE_SUM, SM_SPR and PRE_WIN.) better explain the wildfire occurrence in the nonfire season.

Subsequently, models were constructed using each variable individually to examine the contribution of the variables to the model (Fig. 4b). The results showed that the variables with the highest combined ranking almost always had higher AUC values when constructed individually, and these variables contained information that was more useful for predicting wildfires. In contrast, the variables ranked lower had lower AUC values and made limited contributions to the modeling. The results point to important drivers that are largely consistent with the composite ranking.

Fig. 5 shows the response of wildfire probability to driver changes based on machine learning and the optimum range of the driver that affects wildfire occurrence based on risk detection. They both reveal the relationship between driver variation and wildfire occurrence. For the critical anthropogenic factors of POP and Prox to farmland, their shifts caused highly consistent changes in wildfire probability across seasons, and the range of optimum wildfire occurrence was in general agreement. However, other important seasonal factors, such as PRE_SUM, SM_SPR, and PET_AUT, showed some degree of variation in the response of wildfire probability across seasons, if not the exact opposite response (Fig. 5). For these seasonal drivers, the optimal range of wildfire occurrence was also inconsistent from season to season. For example, the effect of accumulated summer precipitation on wildfire probability in summer, autumn, and winter had a bell shape, but the peak in winter was significantly flatter and inconsistent. The accumulated summer precipitation ranges most likely to trigger wildfires in the four seasons were 69,600–78,800 mm, 76,200–88,600 mm, 67,100–82,000 mm, and 52,900–60,700 mm, respectively (Fig. 5e). Fig. S7 shows the response curves for all other drivers, presenting nonlinearity and variation between seasons.

Table 2
Variable importance of drivers in the fire season and their corresponding rankings per model based on the average rank for each model. Mean decrease accuracy for RF, permutation importance for Maxent and q-values of GeoDetector. Abbreviations same as in Table 1.

Spring wildfire drivers	q-Values (rank)	Permutation importance (rank)	Mean decrease accuracy (rank)	Average rank	Winner wildfire drivers	q-Values (rank)	Permutation importance (rank)	Mean decrease accuracy (rank)	Average rank
POP	0.21 (1)	20.7 (1)	57(1)	1	POP	0.14 (2)	13.1 (2)	25.1 (1)	1.7
Prox to Farmland	0.13 (2)	14.1 (2)	44.1 (2)	2	Prox to Farmland	0.15 (1)	9.5 (4)	24.4 (2)	2.3
NDVI_WIN	0.13 (3)	7.2 (4)	40.7 (5)	4	PRE_WIN	0.11 (4)	18.24 (1)	20.9 (3)	2.7
GDP	0.11 (7)	6.8 (5)	41.7 (3)	5	PET_AUT	0.07 (8)	10.3 (3)	13.5 (8)	6.3
PRE_AUT	0.12 (6)	5.7 (6)	40 (6)	6	GDP	0.14 (3)	1.7 (14)	17.7 (4)	7
PET_AUT	0.13 (4)	4.2 (11)	40.7 (4)	6.3	ELE	0.09 (5)	2.5 (10)	14.2 (7)	7.3
ELE	0.1 (11)	10.2 (3)	32.8 (9)	7.7	ASPECT	0.05 (11)	6.6 (7)	13 (9)	9
SM_SUM	0.11 (8)	3.6 (13)	35.2 (8)	9.7	PRE_SUM	0.08 (6)	1 (16)	14.6 (6)	9.3
PRE_SPR	0.08 (12)	4.4 (9)	30.2 (10)	10.3	SM_WIN	0.05 (13)	8.8 (6)	13 (10)	9.7
SM_SPR	0.12 (5)	0.5 (19)	37.6 (7)	10.3	PRE_SPR	0.07 (7)	1 (17)	15.2 (5)	9.7
RH_AUT	0.07 (13)	4.3 (10)	28.2 (13)	12	WND_AUT	0.06 (10)	3.2 (9)	11.8 (15)	11.3
SM_AUT	0.1 (9)	0.5 (18)	29.5 (11)	12.7	SM_SPR	0.04 (14)	1.9 (13)	12.9 (11)	12.7
WND_SPR	0.06 (16)	5.1 (8)	27.8 (14)	12.7	VPD_SUM	0.03 (20)	9 (5)	11.8 (14)	13
PRE_WIN	0.1 (10)	0.4 (20)	29 (12)	14	SM_AUT	0.04 (16)	2.7 (11)	11.9 (13)	13.3
TWI	0.05 (18)	5.2 (7)	23.4 (17)	14	NDVI_AUT	0.05 (12)	0.8 (18)	12.5 (12)	14
SM_WIN	0.06 (14)	0.9 (15)	27.7 (15)	14.7	SLOPE	0.03 (18)	5.7 (8)	11.6 (16)	14
NDVI_SUM	0.06 (15)	1.1 (14)	25.6 (16)	15	SM_SUM	0.07 (9)	0.5 (19)	11.1 (18)	15.3
SLOPE	0.04 (19)	4 (12)	21.3 (18)	16.3	Prox to Roads	0.04 (15)	2.3 (12)	7.6 (20)	15.7
Prox to Roads	0.05 (17)	0.6 (17)	18.1 (20)	18	RH_SPR	0.04 (17)	1.1 (15)	11.5 (17)	16.3
ASPECT	. (20)	0.8 (16)	19.8 (19)	18.3	TWI	0.03 (19)	0.4 (20)	10 (19)	19.3

Table 3
Variable importance of drivers in the nonfire season and their corresponding rankings per model and based on the average rank for each model. Abbreviations same as in Table 1.

Summer wildfire drivers	q-Values (rank)	Permutation importance (rank)	Mean decrease accuracy (rank)	Average rank	Autumn wildfire drivers	q-Values (rank)	Permutation importance (rank)	Mean decrease accuracy (rank)	Average rank
POP	0.23 (4)	17.3 (1)	27.1 (1)	2	PRE_SPR	0.19 (2)	25.4 (2)	12.2 (1)	1.7
PRE_SUM	0.24 (2)	8.4 (4)	14.9 (4)	3.3	PRE_SUM	0.22 (1)	8 (4)	9.3 (2)	2.3
TMP_WIN	0.29 (1)	2.9 (12)	26.8 (2)	5	SM_SPR	0.18 (3)	0.7 (12)	8.6 (3)	6
NDVI_WIN	0.23 (3)	4.4 (9)	24.1 (3)	5	ELE	0.14 (4)	2 (9)	5.7 (6)	6.3
PET_SUM	0.18 (5)	4.9 (8)	11.7 (5)	6	PRE_WIN	0.12 (7)	2.6 (6)	5.4 (9)	7.3
RH_WIN	0.18 (6)	6.8 (6)	11.2 (7)	6.3	NDVI_SPR	0.13 (5)	0.2 (14)	5.8 (5)	8
PRE_SPR	0.07 (16)	7.1 (5)	11.2 (6)	9	NDVI_SUM	0.12 (8)	2.1 (8)	5.5 (8)	8
GDP	0.15 (9)	1.2 (13)	9.4 (10)	10.7	RH_SPR	0.11 (9)	18 (3)	4.5 (14)	8.7
SLOPE	0.06 (17)	16.6 (2)	7.2 (16)	11.7	SM_SUM	0.12 (6)	0.2 (16)	8 (4)	8.7
SM_SPR	0.16 (7)	0.2 (19)	10.4 (9)	11.7	TWI	– (20)	3.6 (5)	5.5 (7)	10.7
Prox to Farmland	0.1 (14)	6.1 (7)	7.6 (14)	11.7	Prox to Farmland	0.1 (10)	0.4 (13)	5 (10)	11
SM_WIN	0.16 (8)	0 (20)	10.9 (8)	12	VPD_SUM	– (17)	31.2 (1)	3.8 (17)	11.7
WND_SPR	0.11 (13)	3.6 (11)	7.9 (12)	12	ASPECT	– (18)	2.2 (7)	4.7 (11)	12
SM_SUM	0.12 (11)	0.4 (16)	7.6 (13)	13.3	SLOPE	– (19)	1.4 (11)	4.7 (11)	13.7
RH_SUM	0.04 (20)	14.4 (3)	6.3 (18)	13.7	SM_WIN	0.09 (12)	0.2 (15)	4.4 (15)	14
NDVI_SUM	0.12 (12)	0.2 (18)	8.3 (11)	13.7	Prox to Roads	0.09 (13)	1.5 (10)	3.1 (20)	14.3
PRE_WIN	0.14 (10)	0.3 (17)	7.4 (15)	14	POP	0.1 (11)	0 (20)	4.6 (13)	14.7
SM_AUT	0.06 (18)	3.9 (10)	6.5 (17)	15	WND_AUT	– (16)	0.2 (18)	4.6 (12)	15.3
TWI	0.06 (19)	0.9 (14)	6.3 (19)	17.3	SM_AUT	0.08 (14)	0.2 (17)	3.8 (18)	16.3
ASPECT	– (21)	0.4 (15)	5 (20)	18.7	GDP	0.07 (15)	0 (19)	3.4 (19)	17.7
Prox to Roads	0.07 (15)	0 (21)	– (21)	19					

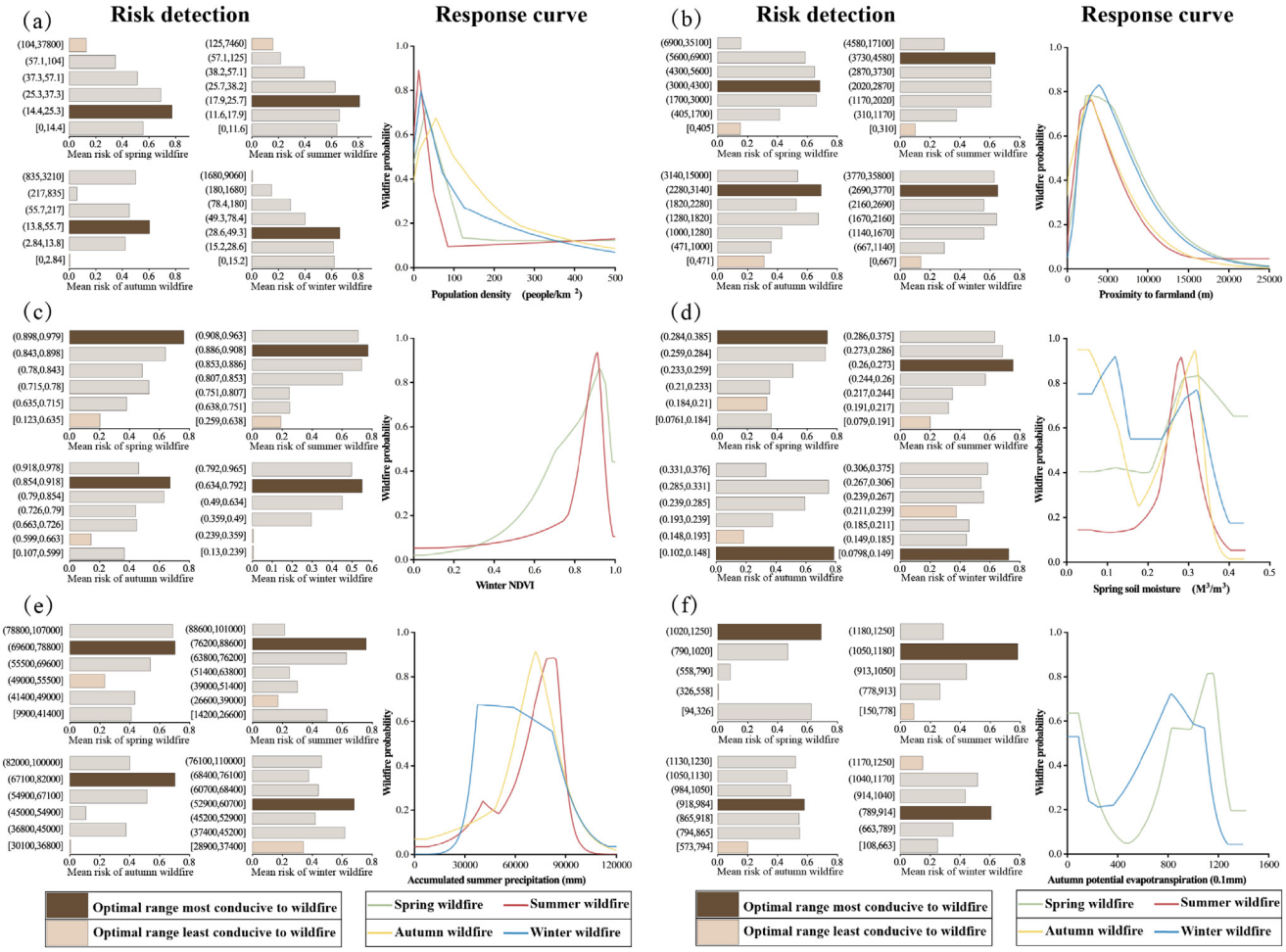


Fig. 5. The relationship between some of the variable drivers and wildfire occurrence. Included is a risk detector (showing the range of drivers corresponding to wildfire risk) and response curves (showing how wildfire probability varies with driver change): (a) POP, (b) Prox to Farmland, (c) NDVI_WIN, (d) SM_SPR, (e) PRE_SUM, (f) PET_AUT. Abbreviations same as in Table 1.

4. Discussion

4.1. Association between drivers and wildfire in different seasons

The seasonal variations in drivers, especially the significant mid-year variations in the NDVI, relative humidity, and potential evapotranspiration, are likely the direct cause of the differences in wildfire distribution across seasons (Fig. S2). We distinguished between seasonal drivers to identify whether changes in these factors would affect wildfire over time. With the exception of summer precipitation and winter precipitation, which clearly dominated wildfires in the current season, the other dominant drivers tended to originate from another season. This result indicates that the dominant drivers tend to be environmental conditions from the previous season or seasons, similar to previous findings (Margolis et al., 2017; Qiu et al., 2022). Although the lag time for drivers to influence wildfire was not specifically indicated, drivers do alter fire behavior on decadal, interannual, and shorter time scales (Crimmins and Comrie, 2004; Yuan et al., 2022). In addition, our findings highlight the importance of climate and precipitation regimes during the dry season. In Yunnan Province, climatic conditions (especially wet conditions) during the dry season and the resulting vegetation changes may be one of the key factors influencing wildfire activity throughout the year. Since the beginning of the 21st century, dry season (winter monsoon period) drought has worsened more in Yunnan Province than in surrounding areas (Zhang et al., 2013). This trend has led to more frequent, intense and continuous dry season droughts, aggravating local wildfire occurrence (Cao et al., 2017).

The mechanism by which seasonal variables affect wildfire is complex, as reflected in the nonlinearity of individual response curves and the lack of overlap between seasons (Fig. 5 and Fig. S7). This may stem on the one hand from the lagged effects that drive the impact of wildfires. On the other hand, it may be attributed to the fact that the effects of seasonal climatic variables on wildfire cannot be reduced to a single outcome but must be considered based on climate, vegetation, and the interactions between them (Flannigan et al., 2009). The above point can also be corroborated by the interaction detection results, i.e., the lower q-values obtained from the single variable and the enhanced effect of their interaction on wildfire (Fig. S3).

In Yunnan Province, the complex mountainous terrain has created a mixed landscape where forests and agricultural lands coexist in a heterogeneous mosaic dominated by agricultural activities that accidentally induced wildfires (Huang et al., 2012). Therefore, as a necessary condition for wildfire initiation, anthropogenic fire sources have a profound impact on wildfires. Our results verify that anthropogenic factors are important variables influencing wildfire occurrence across seasons, especially in spring and winter. The response curve indicates that changes in wildfire probability are related to changes in population density, i.e., the lower the population density is, the higher the likelihood of wildfire, which is consistent with the findings of previous studies (Ma et al., 2022; Sturtevant and Cleland, 2007). Because cities, villages and towns have higher population densities, fires in these places were not considered in this study, which seems to emphasize the role of human activities in extinguishing and suppressing wildfires (Archibald et al., 2012). We also noted that wildfires were more likely to occur relatively close to farmland, which reflects the fact that local wildfires are mainly triggered by human agricultural activities (Benali et al., 2017; Tian et al., 2013). Unlike the seasonal factors, the above anthropogenic variables maintained relative consistency across seasons regarding changes in response curves.

Previous studies have summarized the two driving mechanisms for wildfire initiation. One is related to drought, occurring in vegetated areas where wildfire ignition is not limited by fuel loads but depends on the combustibility of the environment (Abatzoglou and Williams, 2016). The other mechanism pertains to fuel, in which fuel shortages limit wildfire occurrence in unproductive vegetation systems (Pausas and Bradstock, 2007). We suggest that Yunnan Province, which is an area with high vegetation productivity, tends to follow a drought driven pattern in summer and autumn (our results also show precipitation as the primary driver in both seasons) (Ying et al., 2021). However, during spring and winter, the mechanisms driving local wildfire activity may be a combination of drought and fuel related factors. This is when the environment is conducive to the most threatening wildfires due to a sufficient source of combustible material (Pausas and Paula, 2012). Fire sources therefore become critical in determining wildfire distribution during these two seasons (results indicate that anthropogenic activity is the primary driver for both seasons).

4.2. Implications for management

Modeling wildfire probability through different seasonal scenarios can enhance our understanding of the short-term nature of wildfires and may help prevent wildfires. Through modeling, the wildfire susceptibility pattern in Yunnan Province showed a scattered pattern in spring and winter and a concentrated pattern in summer and autumn. Fig. 3 integrates the wildfire susceptibility characteristics throughout the year and reveals potential high wildfire areas. Since wildfire monitoring in Yunnan's rugged terrain is both problematic and expensive, we recommend a seasonal focus on the highly susceptible areas identified in this study for different seasons. Emphasis should be placed on long-term fire prevention efforts in locations where the rate of occurrence remains high throughout the year.

Moreover, strategies for wildfire prevention can be differentiated across seasons. Our findings support the severe measures taken by the local government to cordon off forest areas during the high wildfire season, as this is key to limiting the sources of wildfire. Additionally, in the spring and winter seasons, agricultural and customary fire use should be of utmost concern to the population and managers. In summer and autumn, it is more important to pay attention to the climate, especially the key indicators related to humid conditions, and whether the warning threshold is reached.

Due to the dynamic nature of climate change, the susceptibility pattern of individual seasons is bound to change spatially over time. Additionally, temporal changes, such as the fire season beginning early or late or lasting longer, may be due to the interaction of extreme weather and government policies or a potential shift in the active fire season. Therefore, it is worth further research by regularly updating the results and developing a long-term effective monitoring system for changes in wildfire occurrence both spatially and temporally.

4.3. Uncertainty and limitations

Machine learning was applied to generate, with sufficient accuracy, wildfire susceptibility patterns and model the major drivers for different seasons. However, this study has certain limitations that must be clarified. First, the vegetation factors we considered were relatively homogeneous, focusing only on the live combustible load, which hardly reflects the influence of dead combustible material on wildfire occurrence (Liu et al., 2018; Quan et al., 2021). Although the variables we considered, such as vapor pressure difference and soil moisture, can reflect the flammability of fuels to some extent, they still have limitations. Moreover, we focused on seasonal differences in drivers, but this did not consider anthropogenic differences over time. We assumed that the anthropogenic profile was constant throughout the year; however, it is clear that the season of farming and the rituals of specific festivals (e.g., incense burning and firecrackers) are sources of variation that we did not currently consider (Lin et al., 2008; Martín et al., 2018).

In addition, unlike other regions, straw burning is common in Yunnan Province (Ren et al., 2019). The fires monitored by satellite remote sensing cannot actively remove these planned fires. Some disturbances, such as smoke, may also be incorrectly considered wildfire events. However, we increased the confidence threshold (>80 %) for determining wildfire identification for this reason and excluded fires located in farmland areas where this behavior is prevalent. However, it is still possible that some of the misclassified wildfires were present in our wildfire sample and affected the results. Therefore, refining the criteria for determining wildfire sites in future studies to ensure that satellite fire observations closely match actual wildfires will provide a more reliable sample of dependent variables.

5. Conclusions

We categorized wildfire events into spring, summer, autumn, and winter wildfires based on seasonal dimensions for the 2006–2020 period in Yunnan. Short-term changes in environmental factors were considered to integrate seasonal drivers into the traditional predictor variables. The machine learning model was used to map susceptibility patterns across seasons and assess the contribution and response of drivers to wildfire occurrence to explain the occurrence of wildfires in different seasons. The results showed that the Maxent and RF models achieved good fitting accuracy with similar outcomes. However, the RF modeling results were superior in comparison.

In Yunnan Province, wildfires mostly occur during the spring and winter seasons. The wildfire susceptibility pattern in these two seasons was relatively scattered throughout Yunnan Province, reflecting the high wildfire threat in most areas. Wildfire susceptibility patterns in the other two seasons were more concentrated in one region, with the remaining areas at less potential risk from wildfire during this time. Furthermore, locations in Pu'er had high wildfire susceptibility throughout the year and should be monitored year round.

Among the drivers, the dominant drivers of wildfire occurrence were similar across seasons but not identical. The similarities were that the anthropogenic factors, population density and distance from farmland were considered the more critical dominant factors in all seasons. Differences were found in seasonal climatic variables, with the dominant drivers varying greatly over seasons. Wildfire occurrence was influenced by the climate and vegetation of the current/previous season. Wildfire occurrence in the local nonfire season was driven by drought conditions. However, the fire season consisted of a relatively suitable burning environment and stable fuel supply. Moreover, the fire distribution expressed by anthropogenic variables better reflected the wildfire pattern in the current season. Additionally, the response of wildfire occurrence to the tested variables indicated that the marginal effect of anthropogenic activities on wildfire was generally consistent, while seasonal drivers showed a more varied response curve. Risk detectors supported that seasonal drivers influenced differences in optimal thresholds for wildfire occurrence across seasons. It is concluded that climatic and vegetation factors do not affect wildfire occurrence in the same pattern across seasons, but subsequent studies are needed to delve into the specific associations.

CRediT authorship contribution statement

Wenquan Wang: Conceptualization, Methodology, Investigation, Writing – original draft. **Fengjun Zhao:** Resources, Supervision. **Yanxia Wang:** Formal analysis. **Xiaoyuan Huang:** Data curation. **Jiangxia Ye:** Conceptualization, Writing – review & editing, Project administration.

Data availability

Data will be made available on request.

Declaration of competing interest

All authors declare no potential conflicts of interest with respect to the research, author-ship, and publication of this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2023.161782.

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